# Department of Physics

## National Institute of Technology Kurukshetra

**Course: B.Tech***(CoE, IT, AI&ML, IIOT, M&C, M&V, AI&DS)* **Exam: Sessional-II** **Date: 23-10-2024**

**Subject: Engineering Physics** **Course Code: PHIC 101** **Duration: 50 min**

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| ***Note: All questions are compulsory.*** ***Max. Marks: 20*** ***Roll No:*** | |
| **1.** An electron has a de-Broglie wavelength of $2\times10^{-12}$ *m*. Determine its kinetic energy, as well as the phase and group velocities of its de-Broglie waves. *[Mass of electron= $9.1\times10^{-31}$kg and 1 Joule=$6.24\times10^{18}$ eV.]* | 5 |
| **2.** Draw the following Miller directions [2 1 0], [0 0 1], [$\bar{2}$ 3 $\bar{2}$], [3 $\bar{3}$ 1], and [$\bar{1}$ 2 4]. | 5 |
| **3.** (a) What is Ampere's law? Why Maxwell amended the Ampere's law? Discuss the significance of Maxwell-Ampere's law.<br>(b) A capacitor with circular plates of radius $r=0.05$ m is being charged, and the displacement current between the plates is $2A$. Calculate the magnetic field at a distance $r=0.1$ m from the center of the plates. | 3+2 |
| **4.** (a) At room temperature, the average speed of electrons due to thermal motion is approximately $10^6$ *m/s*. Why, then, don't metals exhibit a corresponding current? How does Drude's theory explain this?<br>(b) A copper wire has a cross-sectional area of $1\ mm^2$ and carries a current of $3A$. If the density of free electrons in copper is $8.5\times10^{28}\ m^{-3}$, calculate the drift velocity of the electrons. | 3+2 |